

Statistical Report

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Statistical Report on Fertility Data in Singapore

Abstract

Introduction

Can time series models trained on data from 1960–2012 accurately forecast Singapore’s continuing fertility decline over 2013–2024?

TO INCLUDE WHEN INTRODUCTION IS DONE

Include necessary abbreviations when writing it up

4. Justify SARIMA period 12 in Introduction (page 3) The periodogram doesn’t obviously point to period 12, so you need literature to justify it. Add a paragraph in the Introduction discussing the Chinese zodiac Dragon year effect on Singapore births. This is what makes the SARIMA(p,d,q)(P,D,Q)[12] models substantively motivated rather than just data-driven.

Methods

The modelling workflow was designed to answer our forecasting needs. The TFR and TLB series were extracted from the dataset, reshaped into annual time series from 1960–2024 and split into a training period from 1960–2012 and a testing period from 2013–2024.

Time plots and the Kwiatkowski-Phillips-Schmidt-Shin (KPSS) tests of the two series were used to check for trends, changes in variance and the transformations required before modelling. Both series showed evidence of non-stationarity and since ARIMA models require the time series to be stationary, transformation of the series was necessary prior to modelling.

The log transformation was used to stabilise variance and make the model proportional rather than absolute changes. Differencing was used to remove the trends shown inside the data.

Both series were then transformed using either just the log or the log with first or second differencing. These transformed series were then tested by the KPSS test to ensure an improvement of the stationarity of the series.

Two transformations showed potential as being stationary (first-differenced and logged, and second-differenced logged). With the KPSS test a small p-value shows evidence against stationarity. For TFR, the KPSS test gave that $p = 0.0406$ for the first-differenced logged series and $p = 0.1$ for the second-differenced logged series. For the logged TLB with both first and second differencing gave the same p-values of $p = 0.1$. So for TLB stationarity was not rejected for either transformation.

Despite the first difference of the TFR data having a KPSS test result below the significance threshold, we would prefer there to be less differencing as we do not want to over-difference the data and introduce noise in a relatively short data-set. We kept the second-differenced models as comparison as well so we could see another set of models in the results.

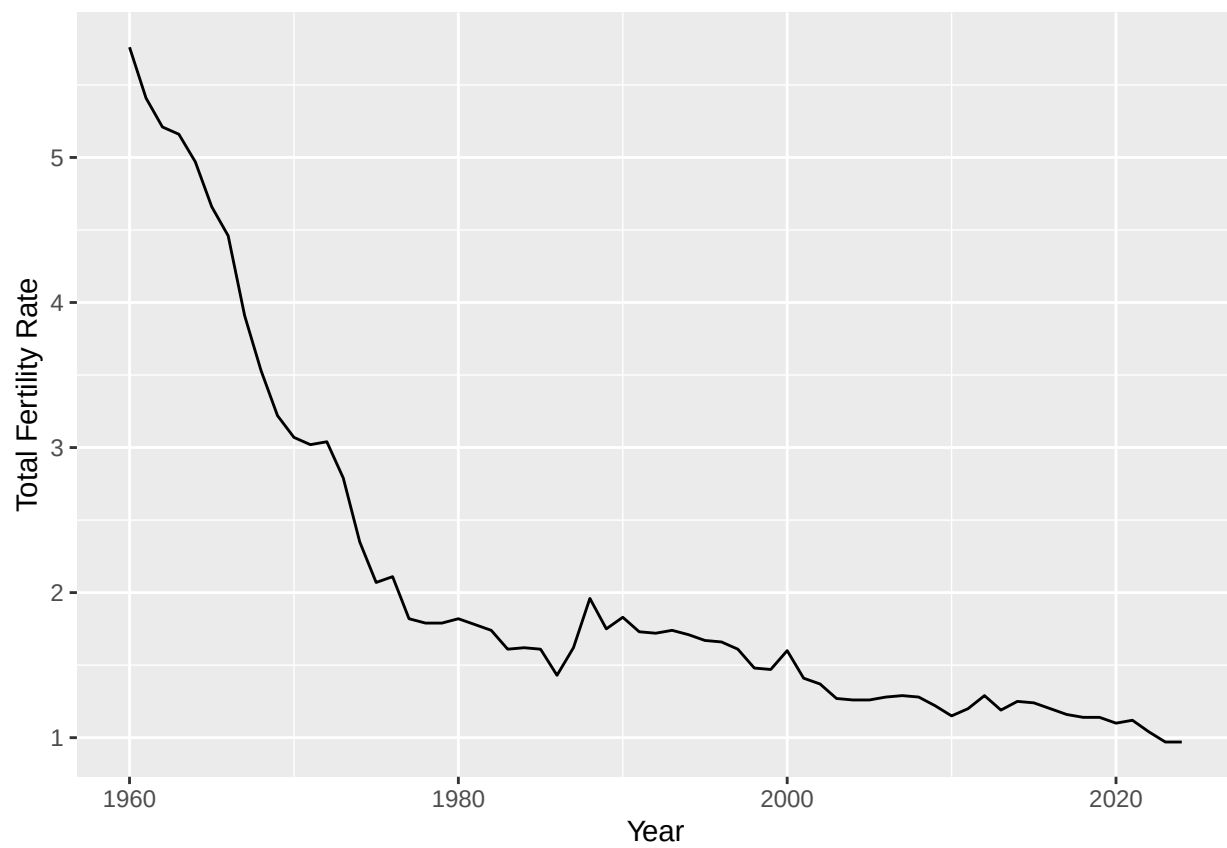


Figure 1: Annual Total Fertility Rate in Singapore

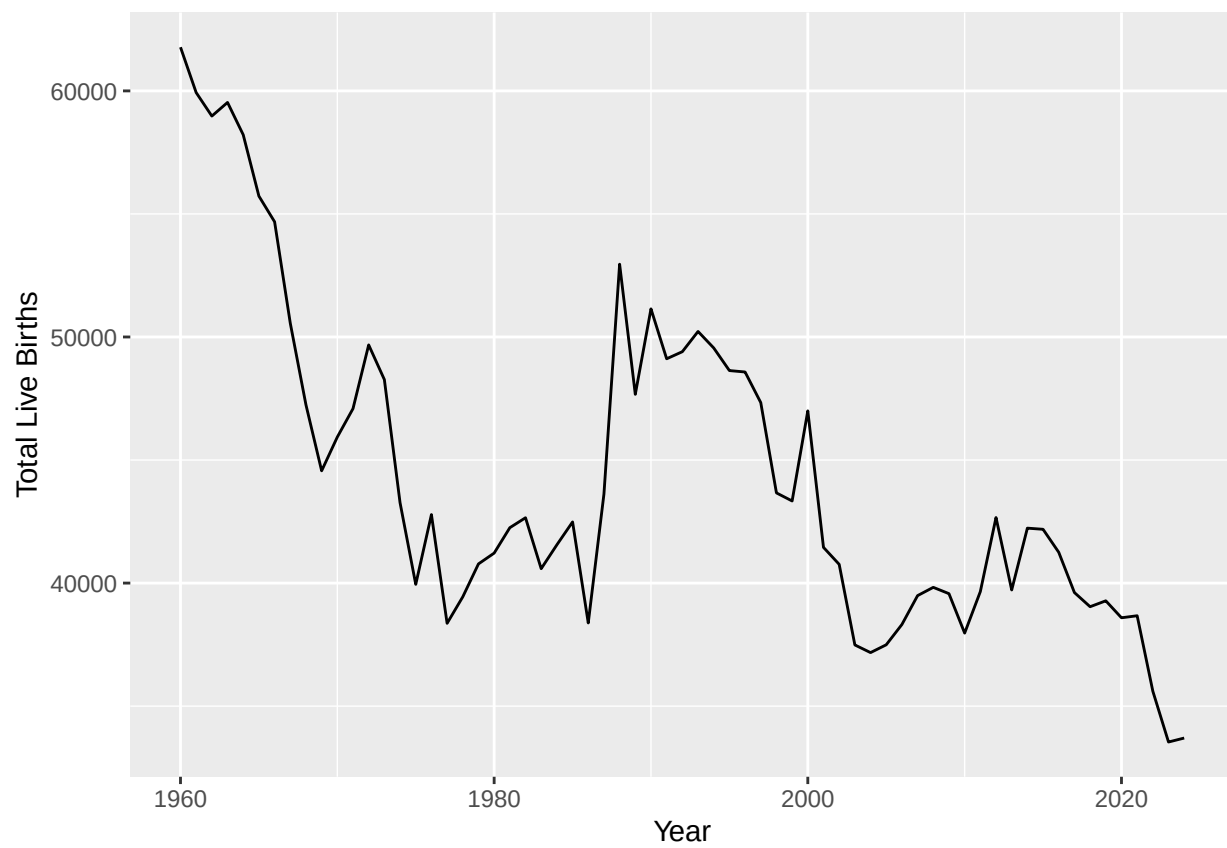


Figure 2: Total Live Births in Singapore

Because the transformed series and baseline residual diagnostics showed dependence around lags 11–13, higher-lag ARIMA models were also considered. These models were included to capture longer-lag autoregressive dependence that was not represented by the low-order baseline models.

SARIMA-style models with period 12 were also considered. Although the data are annual, the period-12 structure was motivated by both the lag-12 dependence observed in the transformed series and the literature discussed in the Introduction on Chinese zodiac-related birth timing effects in Singapore. These models were included to assess whether a 12-year pattern could be represented more simply than by unrestricted high-order ARIMA models. For the SARIMA candidates, the ACF and PACF plots of the transformed series were inspected to guide the initial period-12 model specification.

Models were only considered viable if their residuals were approximately white noise. This again was assessed with ACF and PACF, portmanteau tests and residual plots. Among viable models, the final model selection was based on AIC and forecast accuracy over the 2013-2024 period. Forecast accuracy was assessed with RMSE and MAE, with formulas provided in the appendix.

- So informed decision to try ARMIA (12,x, y) and SARIMA models potentially

Series	Models
TFR	SARIMA(0,1,1)(1,0,0)[12], SARIMA(1,1,1)(1,0,0)[12], SARIMA(1,2,1)(1,0,0)[12]
TLB	SARIMA(0,1,1)(1,0,0)[12], SARIMA(1,1,1)(1,0,0)[12], SARIMA(0,2,1)(1,0,0)[12]

Results

The ACF and PACF plots of the log-first-differenced series are shown in Figure 3. For both TFR and TLB the plots show a remaining dependence on lags 11-13, particularly at lag 12. This suggests that low order ARIMA models may not fully account for the higher order dependencies shown by Figure 3.

To assess the period-12 SARIMA candidates, the ACF and PACF of $(1 - B) \log(X_t)$ (Figure 3) and $(1 - B^{12})(1 - B) \log(X_t)$ (Figure 4) were used to guide the structure of our models. For both series, Figure 3 showed PACF spikes at lag-12 this suggested a seasonal AR term could be useful, while the lower lag ACF structure supported testing simple MA terms. This informed our decision for our first SARIMA candidate model to be SARIMA(0,1,1)(1,0,0)[12]. Nearby candidates were also fitted by varying the non-seasonal AR and MA orders and by comparing seasonal AR and MA terms.

In our seasonally differenced plot (Figure 4), lag-12 dependence remained even after the period 12 differencing. This indicated that differencing alone could not remove the period 12 dependence structure and supported the use of seasonal ARIMA models.

Firstly, we fitted some low-order ARIMA models without seasonal terms, however, as expected based on the lag-12 dependence identified with the ACF/PACF plots these models failed to produce white noise residuals. We next decided to fit some higher-order ARIMA models with seasonal terms as well as some SARIMA models with lag 12. You can see the tested models below with their resultant residuals:

Table 1: Ljung-Box residual tests for TFR candidate models

Model	of Notes
ARIMA(11,1,0)	
ARIMA(12,1,0)	
ARIMA(13,1,0)	
ARIMA(13,1,1)	
SARIMA(0,1,1)(1,0,0)[12]	
SARIMA(0,1,1)(0,0,1)[12]	

Model	of Notes
SARIMA(1,1,1)(1,0,0)[12]	
SARIMA(1,1,1)(0,0,1)[12]	
ARIMA(0,2,1)	
ARIMA(1,2,1)	
SARIMA(0,2,1)(1,0,0)[12]	
SARIMA(1,2,1)(1,0,0)[12]	

Table 2: Ljung-Box residual tests for TLB candidate models

.model	lb_stat	lb_pvalue	viable
tfr_ar11_log	39.061	0.000	Not viable: residual autocorrelation detected
tfr_ar12_log	30.821	0.002	Not viable: residual autocorrelation detected
tfr_ar13_log	22.633	0.020	Not viable: residual autocorrelation detected
tfr_ar13ma1_log	20.852	0.022	Not viable: residual autocorrelation detected
tfr_sar_011_100_log	30.867	0.099	Viable: residual autocorrelation not detected
tfr_sar_011_001_log	36.434	0.027	Not viable: residual autocorrelation detected
tfr_sar_111_100_log	29.273	0.108	Viable: residual autocorrelation not detected
tfr_sar_111_001_log	41.971	0.004	Not viable: residual autocorrelation detected
tfr_021_log	44.300	0.005	Not viable: residual autocorrelation detected
tfr_121_log	44.296	0.003	Not viable: residual autocorrelation detected
tfr_sar_021_100_log	43.603	0.004	Not viable: residual autocorrelation detected
tfr_sar_121_100_log	30.536	0.082	Viable: residual autocorrelation not detected

.model	lb_stat	lb_pvalue	viable
tlb_ar11_log	36.246	0.001	Not viable: residual autocorrelation detected
tlb_ar12_log	25.705	0.012	Not viable: residual autocorrelation detected
tlb_ar13_log	20.023	0.045	Not viable: residual autocorrelation detected
tlb_ar13ma1_log	15.689	0.109	Viable: residual autocorrelation not detected
tlb_sar_011_100_log	29.426	0.133	Viable: residual autocorrelation not detected
tlb_sar_011_001_log	36.300	0.028	Not viable: residual autocorrelation detected
tlb_sar_111_100_log	28.953	0.115	Viable: residual autocorrelation not detected
tlb_sar_111_001_log	36.603	0.019	Not viable: residual autocorrelation detected
tlb_021_log	45.732	0.003	Not viable: residual autocorrelation detected
tlb_121_log	43.206	0.004	Not viable: residual autocorrelation detected
tlb_sar_021_100_log	34.005	0.049	Not viable: residual autocorrelation detected
tlb_sar_121_100_log	28.699	0.121	Viable: residual autocorrelation not detected

We then tested each of these models

Your professor expects the Results section to show:

ACF/PACF plots of the detrended and deseasonalised series
How the first viable model was identified from those plots
How you varied parameters to get additional viable models
How each viable model fits the testing data

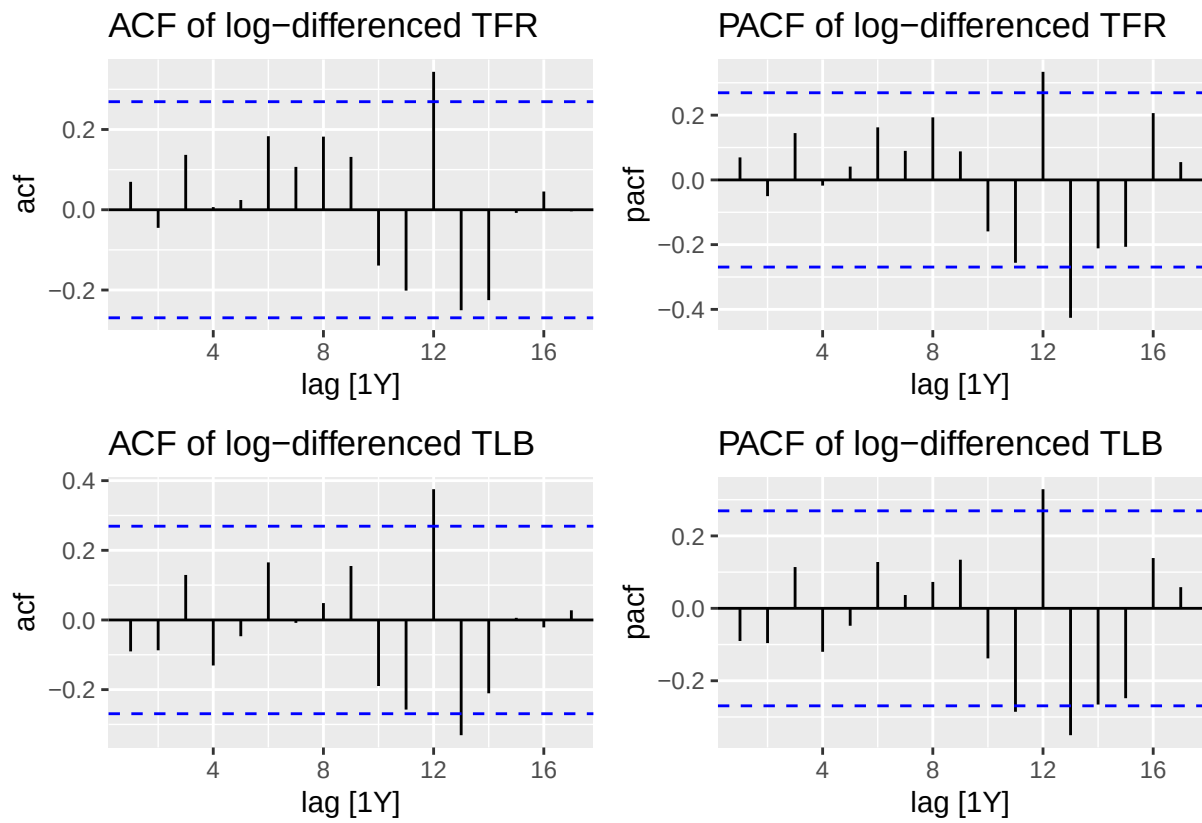


Figure 3: ACF and PACF plots of the log-first-differenced TFR and TLB series

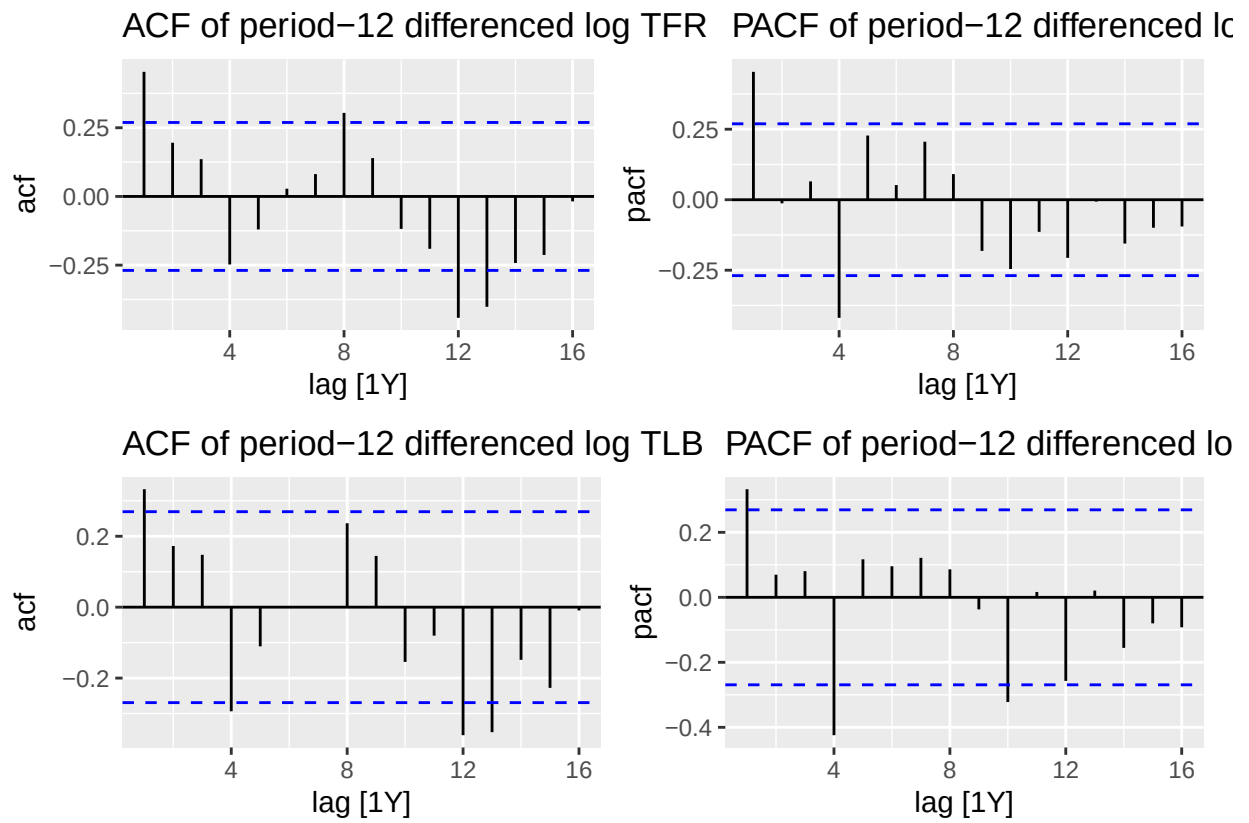


Figure 4: ACF and PACF plots of the log-first-differenced and period-12 differenced TFR and TLB series

Discussion

Conclusion

References

Appendices if applicable.